

Python Practice Handout

Strings and f-strings

1. What is a string

A string is how Python represents text. You have already used numbers with int and float. A string is the third type, and it holds words instead of numbers.

```
shirt = "blue"
```

2. Strings need quotes

Text must have quotes around it. Without them Python thinks blue is a variable name, and it has not been created, so you get an error.

This does not work:

```
shirt = blue
```

Error:

```
NameError: name 'blue' is not defined
```

This works:

```
shirt = "blue"
```

3. Printing a string

Nothing appears until you print it. Notice that the quotes do not show up in the output. They only tell Python where the text begins and ends.

```
shirt = "blue"  
  
print(shirt)
```

Output:

```
blue
```

4. Single or double quotes

You can use double quotes or single quotes. Both give exactly the same result. The one thing you cannot do is start with one kind and end with the other.

```
shirt = "blue"  
shirt = 'blue'
```

This is not allowed:

```
shirt = "blue'
```

5. Quotes inside a string

If your text contains an apostrophe, single quotes around it will break the string early. Wrapping it in double quotes instead fixes it.

This does not work:

```
store = 'Nick's Pizza Shop'
```

This works:

```
store = "Nick's Pizza Shop"
```

6. Using a backslash

Sometimes the text needs both kinds of quote inside it. Put a backslash in front of a quote to tell Python it is part of the text, not the end of the string.

```
store = 'Nick\'s Pizza Shop, the "best" there is'  
print(store)
```

Output:

```
Nick's Pizza Shop, the "best" there is
```

7. Try it yourself

Put the name of a movie you like into a variable called movie. Do not forget the quotes. Try this on your own before looking below.

```
movie = 'Sandlot'  
print(movie)
```

8. Variables inside a string

Say you have some variables and you want to write a sentence about them. If you type the values into the string by hand, the sentence never changes when the variables change.

```
day = 21  
month = "October"  
temp = 65  
  
print("Today is October 21 and it is 65 degrees outside")
```

Change day to 25 and the sentence still says 21. The text was typed in, so it is stuck.

9. f-strings

To put a variable inside a string, add a lowercase f just before the opening quote. Then wrap any variable name in curly brackets and Python fills in its value.

```
day = 25

print(f"Today is October {day} and it is 65 degrees outside")
```

Output:

```
Today is October 25 and it is 65 degrees outside
```

10. Using several variables

You can put as many variables as you like inside one f-string. Ints, floats and strings all work.

```
day = 3
month = "September"
temp = -15

print(f"Today is {month} {day} and it is {temp} degrees outside")
```

Output:

```
Today is September 3 and it is -15 degrees outside
```

Now change any variable and run again. The sentence updates on its own.

11. Names with more than one word

When a variable name needs two words, put an underscore between them. It is much easier to read. This is normal practice in Python.

```
day_name = "Tuesday"
```

12. Try it yourself

Make a variable called `day_name`, set it to a day of the week, and add it to the sentence above. Try this on your own before looking below.

```
day = 3
month = "September"
temp = -15
day_name = "Tuesday"

print(f"Today is {day_name}, {month} {day} and it is {temp} degrees outside")
```

Output:

```
Today is Tuesday, September 3 and it is -15 degrees outside
```